

ROOP SAGAR MANGINENI

Machine Learning Engineer | Software Engineer

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 3+ years of experience building scalable ML and NLP systems across enterprise and research environments. Proven expertise in Python, PyTorch, TensorFlow, and cloud-based ML deployment on AWS. Experienced in designing end-to-end ML pipelines, production of APIs, and data processing systems that improve model accuracy, automate workflows, and support real-time inference. Strong background in LLMs, NLP systems, and scalable backend architecture.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, C++

Machine Learning: Scikit-learn, TensorFlow, PyTorch, NLP, LLMs, BERT, Feature Engineering, Model Optimization

ML Systems & Data Engineering: Data Pipelines, ETL, Pandas, NumPy, Data Processing, Feature Stores

Backend Development: Flask, FastAPI, REST APIs, Microservices

Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Git, GitHub Actions, CI/CD

Databases: PostgreSQL, MySQL, Snowflake

Tools: Linux, Streamlit, Retool

Core Concepts: Machine Learning Systems, System Design, Distributed Systems, Data Structures

PROFESSIONAL EXPERIENCE

Prosrvc LLC – USA

Software Engineer (Machine Learning)

May 2025 – Present

- Designed and deployed machine learning models for device resale price prediction and hardware defect detection, achieving 87%+ prediction accuracy and improving automated decision-making.
- Built scalable data processing pipelines using Python, Pandas, and SQL to process 10K+ device records monthly, improving data reliability for ML training and analytics.
- Developed production-grade REST APIs using Flask to serve real-time ML inference, enabling seamless integration of predictive models into internal business systems.
- Implemented model validation, logging, and monitoring frameworks, improving reliability and observability of production ML services.
- Collaborated with cross-functional teams to integrate ML models into backend applications and analytics dashboards, improving operational efficiency and data-driven insights.

University of Illinois – USA

Graduate Research Assistant (Machine Learning / NLP)

Aug 2024 – Apr 2025

- Optimized BERT-based NLP models, improving classification accuracy from 13% to 86.4% through advanced feature engineering and hyperparameter tuning.
- Designed large-scale NLP preprocessing pipelines using Python, improving training dataset quality and model performance.
- Developed semantic similarity and document matching algorithms for automated text classification and information retrieval.
- Conducted model evaluation using precision, recall, F1-score, and accuracy metrics, improving predictive performance and model robustness.
- Collaborated with research teams to develop scalable ML experiments and reproducible pipelines for academic research and model benchmarking.

Wipro – India

Machine Learning Engineer

Apr 2022 – Dec 2023

- Developed and optimized machine learning models for enterprise applications using Python and Scikit-learn, improving prediction accuracy and automation.
- Built automated data preprocessing and feature engineering pipelines, reducing manual data preparation time and improving ML workflow efficiency.

- Processed large-scale datasets using Pandas, NumPy, and SQL, enabling efficient data transformation and feature extraction for ML training.
- Collaborated with software engineering teams to deploy ML models into production systems using REST APIs and backend services.
- Improved model performance through feature selection, hyperparameter tuning, and model evaluation techniques.

PROJECTS

Job Search Web App | [Link](#) | Technologies: Python, Streamlit, NLP, Rapid API, Pandas

- Built an NLP-based job recommendation system achieving 90%+ skill extraction accuracy.
- Integrated external APIs to automatically collect and process job listings.
- Implemented text similarity algorithms to improve job matching relevance.
- Integrated CI/CD pipelines using Jenkins and GitHub Actions for automated deployment.
- Improved system scalability supporting 30% higher user traffic.

Automated Blog Generation using Llama 2 | [Link](#) | Technologies: Python, LangChain, LLM, Streamlit, GenAI

- Built an AI-powered blog generation platform using Llama-2 and LangChain.
- Developed modular REST APIs for content generation pipelines.
- Reduced manual content creation time by 85% using generative AI workflows.
- Designed prompt engineering strategies to improve generated content quality.
- Optimized backend performance by reducing API response time by 25%.

EDUCATION

Master of Science in Computer Science University of Illinois	Jan 2024 – May 2025 USA
Bachelor of Technology in Computer Science SRM University	July 2019 – June 2023 INDIA

CERTIFICATIONS

Google Generative AI Leader | [Link](#) | 2026 – 2029

ACHIEVEMENTS

- Selected as Graduate Research Assistant among 150+ candidates at University of Illinois.
- 1st Prize – Hack SRM Hackathon for building an innovative scalable solution.
- 3rd Prize – Top Hat Hackers Hackathon for workplace accessibility technology.
- President – Data Science Club (SRM AP) leading technical workshops and ML initiatives.